Question	Answer	Marks	Partial Marks
5(b)	14	**2**	**M1** for $12-\frac{k}{2}=5$ or $2^{\frac{k}{2}}=\frac{2^{12}}{2^5}$ oe or $\frac{4096}{32}$ or $12-5$ or $2^{12}\div 2^{\frac{14}{2}}$ [= 32] seen
5(c)	$2y^3-3y^2-23y+12$ final answer	**3**	**B2** for correct unsimplified expanded expression or for simplified four-term expression of correct form with 3 terms correct or **B1** for correct expansion of 2 of the brackets with at least 3 terms correct
5(d)	$[x=]\frac{3}{y-1}$ final answer	**3**	**M1** for $xy = 3 + x$ **M1** for $xy - x = 3$ or $x-\frac{x}{y}=\frac{3}{y}$ **M1** for factorising and dividing
6(a)(i)	$\frac{1}{3}$ oe	**1**	
6(a)(ii)	100	**1**	**FT** *their* **(a)(i)** × 300 to at least 3 sf or rounded to the nearest integer
6(b)(i)	$\frac{2}{15}$ oe	**3**	**M2** for $4\times\frac{1}{6}\times\frac{1}{5}$ oe or **M1** for $k\left(\frac{1}{6}\times\frac{1}{5}\right)$ oe or list or indication of 4 correct pairs
6(b)(ii)	$\frac{3}{5}$ oe	**3**	**M2** for $1-\frac{4}{6}\times\frac{3}{5}$ or $2\left(\frac{2}{6}\times\frac{4}{5}\right)+\frac{2}{6}\times\frac{1}{5}$ oe or $\frac{2}{6}+\left(\frac{4}{6}\times\frac{2}{5}\right)$ oe or **M1** for $\frac{4}{6}\times\frac{3}{5}$ oe seen or $\frac{2}{6}\times\frac{4}{5}[\times 2]$ oe seen or $\frac{2}{6}\times\frac{1}{5}$ oe seen or correct identification of 18 pairs or space diagram oe
7(a)	$n - 5 + 3n + 10 > 105$ or better	**B1**	
	$n > 25$ final answer	**B2**	**M1** for $4n > 100$

Question	Answer	Marks	Partial Marks
7(b)	4.8	**3**	**M1** for $y = \frac{k}{x^2}$ or better **M1** for $[y =]\frac{their\ k}{5^2}$ OR **M2** for $y \times 5^2 = 7.5 \times 4^2$
7(c)(i)	$6 - 2n$ oe final answer	**2**	**B1** for answer $6 - kn$ $(k \neq 0)$ oe or answer $j - 2n$ oe or for correct expression shown in working and then spoilt
7(c)(ii)	$2n^2 - 1$ oe final answer	**2**	**B1** for 2nd diff = 4 or a quadratic expression or for correct expression shown in working and then spoilt
8(a)(i)	2.67 or 2.666…	**3**	**M2** for $\frac{6 \times \sin 25}{\sin 72}$ or **M1** for implicit version
8(a)(ii)	4.14 or 4.140…	**3**	**M1** for $6^2 + 7.4^2 - 2 \times 6 \times 7.4 \times \cos 34$ **A1** for 17.1 to 17.2
8(a)(iii)	20.4 or 20.35 to 20.36…	**4**	**B1** for angle *SQR* = 83 **M1** for $\frac{1}{2} \times 6 \times$ *their* **(a)(i)** $\times \sin$ *their* (180–72–25) oe **M1** for $\frac{1}{2} \times 6 \times 7.4 \times \sin 34$ oe
8(b)(i)	8.7[0] or 8.695…	**4**	**B3** for $\sqrt{980}$ oe or 31.3 or 31.30… or **M3** for $40 - \sqrt{20^2 + 18^2 + 16^2}$ oe or **M2** for $20^2 + 18^2 + 16^2$ oe or **M1** for any correct attempt at 2-dimensional Pythagoras' e.g. $18^2 + 16^2$
8(b)(ii)	30.7 or 30.73 to 30.74…	**3**	**M2** for $[\sin =]\frac{16}{\sqrt{20^2 + 18^2 + 16^2}}$ oe or **B1** for identifying angle *GAC*